HIRESIGHTS

Rakamin

Data-Driven Recruitment System

Optimizing Biased Talent Acquisition Through Data-Driven HR Analytics

Business Understanding

Business & Industry Research

The Strategic Context

The Paradigm Shift

HR management is evolving from qualitative "gut-feelings" to objective, logical People Analytics.

Talent as a Competitive Asset

Data-driven organizations are 19x more likely to be profitable than non-data-driven peers.

Scale of Engagement

Manual screening is not possible on large applicant volumes (e.g. Google ~2M applications) without sacrificing quality.

Market Trends (2024–2026)

Market Projections

Global spending projections for AI & analytics in HR by 2025 with a compound annual growth rate of 24.5%.

68% Adoption

Companies today are already using data-driven recruitment strategies for efficiency.

44% Reskilling

Focus shifts from past positions to validated skill sets. The World Economic Forum 2025 of workers will require reskilling by 2030.

Problem & Goal Statements

The Risks of Conventional Hiring

The Cost of Failure

Harvard Business Review reports that 80% of employee turnover is caused by poor hiring decisions.

Financial Leakage

A single "bad hire" can cost an organization between 50% and 200% of the employee's annual salary when factoring in recruitment, onboarding, and lost productivity. Assuming the annual salary of a tech employee is ~IDR 63.36 million, an organization might lose 31.68 to 126.72 million rupiah.

Subjectivity Bias

HR team struggles to make consistent and fair hiring decisions due to reliance on subjective judgment. This leads to inefficiency, bias, and missed opportunities.

Goals & Objectives

Goals

Transform the conventional recruitment process into a data-driven system to improve talent selection accuracy and company operational cost efficiency.

Objectives

Build an objective, automated model by analyzing qualification variable correlations, determining success score thresholds, and conducting bias audits to ensure fairness in recruitment decision-making.

Success Metrics & Business Impact

The KPIs for Data-Driven Era

F1-Score

The main benchmark for balancing Precision and Recall.

Quality of Hire (QoH)

Measured by the 15% average improvement in how well new employees fit their roles and contribute to company goals.

Hiring Efficiency

Reduce manual screening by up to 80%.

Selection Accuracy

Machine learning models (e.g., Random Forest, XGBoost) can achieve accuracy rates exceeding 95% in predicting candidate suitability based on historical high-performer data.

The Value Proposition

1. Operational Cost Efficiency

Organizations adopting data-driven strategies report a 20-30% reduction in hiring costs by decreasing reliance on third-party agencies and manual labor.

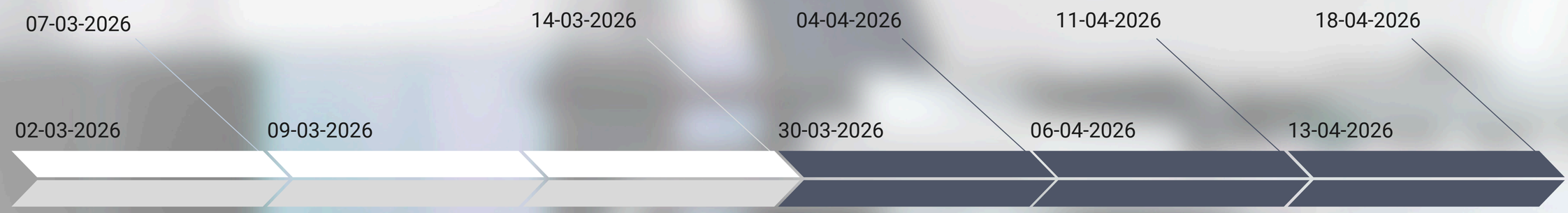
2. Reduced Attrition

Predictive models can identify high-risk departures with 90% accuracy, allowing for proactive interventions that protect your recruitment ROI.

3. Diversity & Inclusion

Data-driven systems can reduce unconscious bias by anonymizing candidate profiles (removing name, gender, age) and focusing strictly on objective qualifications.

Project Timeline & Workflow (CRISP-DM)



Stage 0
Business Understanding
Problem Statement, Goals & Objectives

Stage 1
Data Understanding
Exploratory Data Analysis

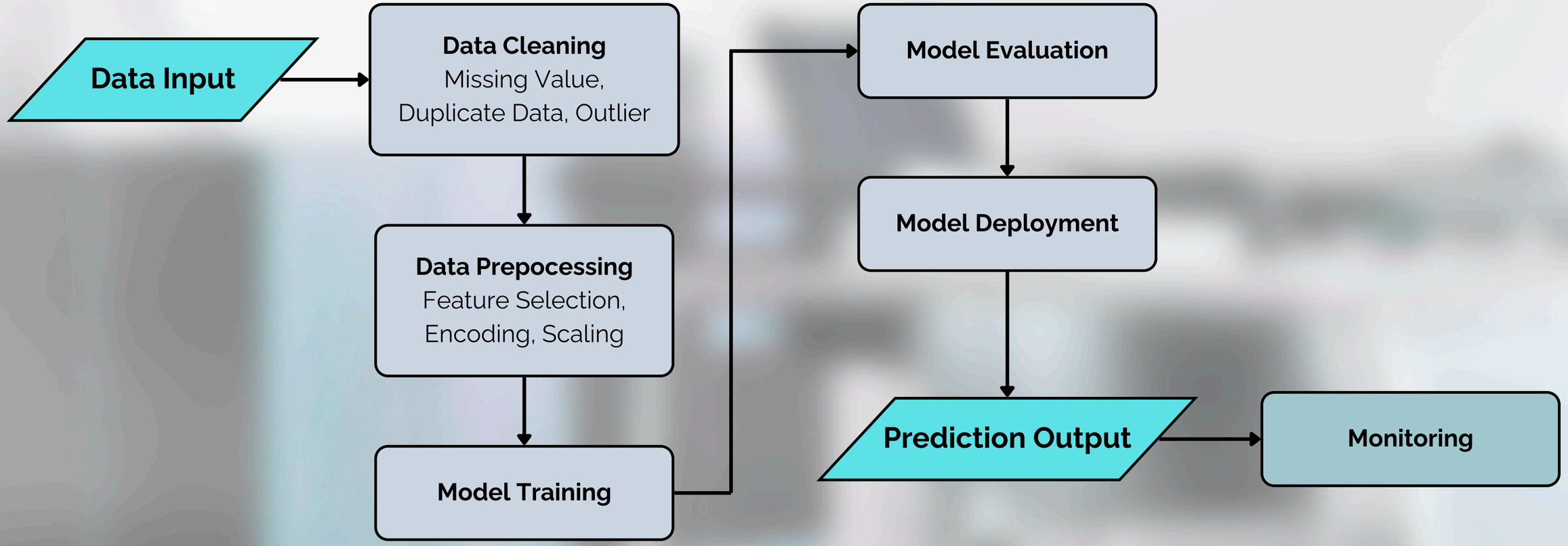
Data Preparation
Preprocessing & Feature Enginering

Stage 2
Modeling
Model Training & Prediction

Stage 3
Model Evaluation
Performance Assessment & Validation

Stage 4
Deployment
Integration & Release

Pipeline Model



Risk-Feasibility

Aspect	Potential Risk	Mitigation Strategy	Feasibility
Data	Historical bias in the 1,500 candidate records due to previous subjective manual hiring decisions.	Conduct regular Fairness & Bias Audits.	Data is available for auditing to ensure objective outcomes.
Technical	Model fails to balance selection precision with the ability to capture potential talent.	Use **F1-Score** to balance Precision and Recall to reduce bias in modeling, hence minimize bad candidate and maximize good candidate acceptance.	F1-Score provides a robust technical validation framework.
Operational	HR teams find it difficult to adopt new systems.	System integration that provides instant predictions.	Data-driven systems minimize intuition-based risks.
Economic	Marketing budget wasted on recruitment channels with low conversion rates.	Perform data-driven evaluations of **Recruitment Strategy ROI** to optimize budget allocation.	ROI analysis ensures targeted and cost-effective spending.
Strategic	Inaccurate threshold settings for candidate scores, leading to high-risk hires.	Execute **Sensitivity Analysis** to identify critical score thresholds that guarantee >80% hiring success.	Ensures only top-quality talent is onboarded to boost productivity.